

TO: ME 2171 Instrumentation Lab Instructional Staff
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SUBJECT: Lab 3, Analog-to-Digital & Digital-to-Analog

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1.0 Statement of Purpose

The objectives for this lab included utilizing an oscilloscope and the function generator and obtaining the analog signal from the oscilloscope and obtaining the data on LABVIEW via the use of the A/D board. Using the A/D board, we were both able to send digital input to analog, or gather digital output from analog. From the data we could find what the amplification factor was defaulted to.

2.0 Background

The model number for the A/D converter (DAQ board) was the NI USB-6221 with 8 analog input channels and 2 analog output channel ports. The DAQ board is functional for both receiving analog input and sending out digital input to analog. When the DAQ board is converting from analog to digital, it uses the multiplexer and other components. The board includes an analog to digital converter multiplexer that switches between channels. Data goes through the DAQ's multiplexer and also passes through an amplifier (which is set automatically) with a certain gain factor. The value of this gain factor is set by the DAQ board. After amplifying the signal, it is then converted from analog voltage into 16-bit count value in binary with the A/D converter, which has a set sampling rate. For this lab we have set the sampling rate to a fixed 200 kHz. The DAQ also includes a digital to analog converter where the digital values(binary) are converted to analog voltages in where the board sends these voltages to the analog channels.

3.0 Discussion

3.1 Equipment

- LabVIEW (computer that runs LabVIEW)
- NI-USB-6221 (DAQ) board
- Siglent SDG 1022XPlus Function Generator
- Tektronix 2-series mixed signal oscilloscope
- 2 BNC cables
- BNC T-connector

3.2 Sampling an Analog Input

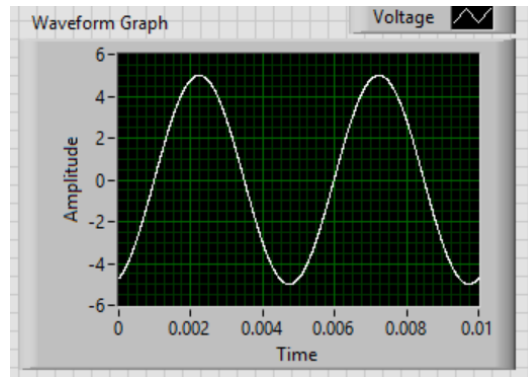


Figure 1 0.2 kHz sampled waveform (time range 0-0.01)

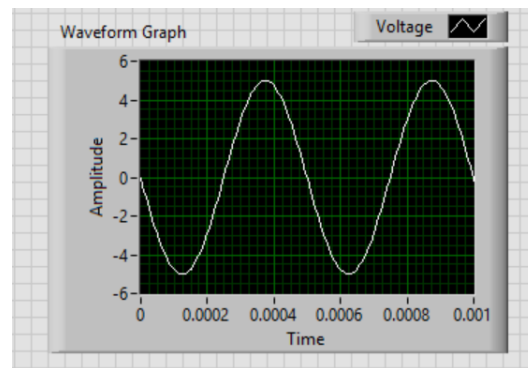


Figure 2 2kHz sampled waveform (time range 0-0.001).

Fig 1. and Fig 2. both showed similar outputs as what was displayed on the oscilloscope. Our team observed both values for Fig. 1 and noted that both, the frequency on the waveform chart and the frequency on the oscilloscope, were identical. This likely meant that Fig 1. had transposed the input signals without aliasing.

Fig 2. is a similar case in where we are displaying a larger input signal, but we did not change the sample rate. The sample rate for Fig 1. and Fig 2. were set to 200kHz. Since our frequency was increased by a factor of 10, it should be noted that our Fig 2. also shows the x-window settings on the front panel modified from 0-0.01 to 0-0.001, which is why the waves look identical to Fig 1, however, our frequency for Fig 2. is happening at a rate 10x more than Fig 1. The wavelength and period are still matching the wavelength and period respectively from the oscilloscope, as both have changed window settings to accommodate for the 10x larger frequency.

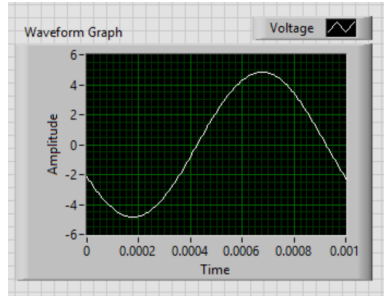


Figure 3. 201 kHz sampled waveform (time range 0-0.001)

Lastly, Fig 3. was configured to display a 201 kHz signal that is still being sampled at 200kHz on the front panel's waveform chart. In this case our waveform is no longer identical to the oscilloscope's display of the signal. This indicates that we are gathering part of the signal, which results in an aliased frequency. Due to our frequency for Fig 3. undergo aliasing, our waveform chart appears to have a longer period that is not consistent with oscilloscope, nor the frequency on the function generator.

In terms of the Nyquist criterion, we can compare the sampling rate by twice of each signal frequency and compare if the sampling rate is larger. If the sampling rate is larger, then Nyquist criterion is followed. If the sampling rate is less than twice the input signal, not all of the frequency is being converted correctly, and the Nyquist criterion is not followed. For each case, the calculations are shown below:

$$\text{Eq 1). } 200 \text{ Hz} \times 2 < 200 \times 10^3 \text{ Hz}$$

$$\text{Eq 2). } 2 \text{ kHz} \times 2 < 200 \text{ kHz}$$

$$\text{Eq 3). } 201 \text{ kHz} \times 2 > 200 \text{ kHz}$$

From each equation we can see Eq 1. & Eq 2. follow the Nyquist criterion, while Eq 3. violates it as the input signal*2 is equal to 402kHz, which is significantly larger than the sampling rate. The Nyquist criterion being violated signifies that the sampling rate is too little for the given frequency, which results in aliasing. This explains why when looking at Fig 3. In comparison to Fig 2., Fig 3. appears to be a lower frequency. If aliasing occurs, the signal is not completely converted correctly, resulting in a different chart from the input.

If all 8 analog input channels are being checked by the multiplexer, then, the switching between channels would have to be accounted for in the calculations in the Nyquist criterion (in previous calculations the multiplier for the multiplexer channels was just 1). This would mean the frequency calculations would be $(\frac{200\text{kHz}}{8 \text{ channels}} * \frac{1}{2}) = 12.5\text{kHz}$. Therefore 12 kHz would be the max input frequency without resulting in aliasing.

3.3 Transfer functions for two A/D board input ranges

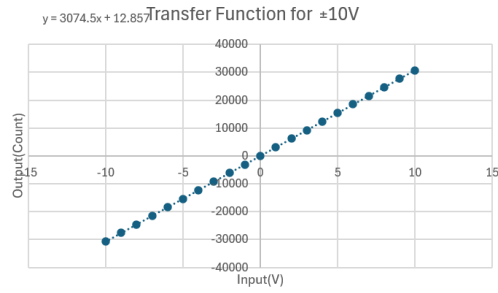


Figure 4 Transfer function $\pm 10V$

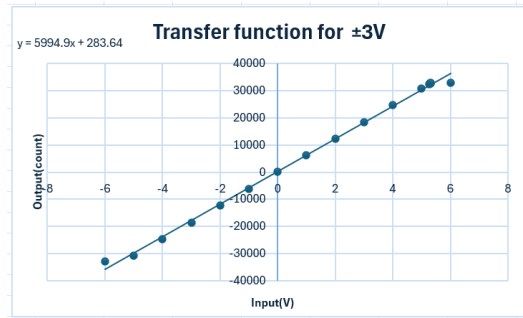


Figure 5. Transfer function $\pm 3V$ without saturation data points

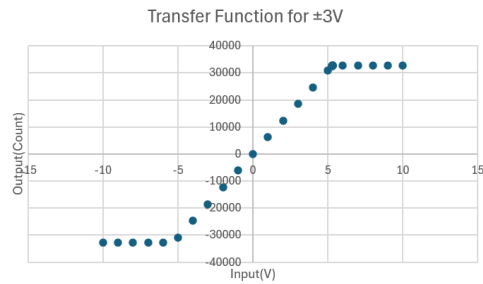


Figure 6. Transfer function $\pm 3V$ with all data points

Input	output	input(V)	output
-10	-30707	-10	-32768
-9	-27646	-9	-32768
-8	-24582	-8	-32768
-7	-21520	-7	-32768
-6	-18450	-6	-32768
-5	-15384	-5	-30839
-4	-12306	-4	-24677
-3	-9229	-3	-18512
-2	-6149	-2	-12346
-1	-3056	-1	-6146
0	23	0	23
1	3104	1	6192
2	6170	2	12333
3	9249	3	18497
4	12328	4	24669
5	15403	5	30830
6	18476	5.27	32488
7	21540	5.28	32551
8	24606	5.29	32612
9	27669	5.3	32671
10	30731	5.31	32733
		5.32	32767
		6	32768
		7	32768
		8	32768
		9	32768
		10	32768

Table 1. Transfer function data points for $\pm 10V$ & $\pm 3V$

For the $\pm 10V$ and the $\pm 3V$ transfer functions, the $\pm 10V$ had a full range from $\pm 10V$, so we can simply twice the highest binary number. Although this would not be accurate as there may be more binary points. Consequently, the $\pm 3V$ had a smaller range from $\sim \pm 6V$, where points $> 6V$ resulted in saturation, with a binary count value of ± 32768 , as seen from both Table 1. And Fig 4. To calculate the number of bits we can see the highest binary count for each case and multiply it by 2 in order to account for both positive and negative counts. For $\pm 10V$ we get 65536 counts and for $\pm 3V$ we obtain 65536 counts. For each of these cases, this makes sense as the settings are set early on as “2D|16”, indicating a 16-bit setting. This means we would have $2^{16} = 65536$ counts on the A/D board. Knowing that binary is a 2 number system, we could also take $\log_2 65536 = 16$, meaning we are working with 16 bits.

To determine the amplification factor of the calibration curve for both transfer functions we can see that the count for the $\pm 10V$ is set to a factor of 1 as there is no saturation indicated within the range and so each voltage value has a unique count ($\pm 10V/10V = 1$). For the Transfer function for $\pm 3V$ range, we can see that saturation ranges from $\pm 6V$. From this we can calculate the gain factor by dividing the range at which saturation ranges by our input ranges, $6V/3V = 2$. This means our amplification factor for the second transfer function is 2.

Calculating static sensitivity and resolution can be found from the slope of each transfer function and from calculating the resolution by taking the inverse. We use two points from each transfer function to work as the max counts on each transfer function. We then divide by the input ranges and then multiply by the gain. We get values of 3071.9 Counts/V (for the $\pm 10V$ transfer function) and 6168.2 Counts/V (for the $\pm 3V$ transfer function).

$$\text{Static Sensitivity: } \frac{(30731 - (-30707))}{10 - (-10)} = 3071.9 \text{ Counts/Volt}$$

$$\text{Resolution: } \frac{1V}{3071.9 \text{ Counts}} = 0.000325 \frac{V}{\text{Count}} = 3.25 \times 10^{-4} \frac{V}{\text{Count}}$$

And for the $\pm 3V$ transfer function:

$$\text{Static Sensitivity: } \frac{(18497 - (-18512))}{3 - (-3)} = 6168.2 \text{ Counts/Volt}$$

$$\text{Resolution: } \frac{1V}{6168.2 \text{ Counts}} = 0.000162 \frac{V}{\text{Count}} = 1.62 \times 10^{-4} \frac{V}{\text{Count}}$$

From these values we can see that when we decreased the input range from $\pm 10V$ to $\pm 3V$, the resolution gets smaller (which means a finer or increased resolution) and

static sensitivity increases. Therefore, by decreasing the input range, we have a nicer and finer resolution.

To obtain the ideal amplification factor for the finest resolution of the $\pm 3\text{V}$ input signal, we would need an amplification factor that would begin to saturate right before $\pm 3\text{V}$, which would result in the finest resolution as it would give us the smallest Volts per Count without limiting the range inside $\pm 3\text{V}$. Therefore, by dividing $10\text{V}/3\text{V}$, we can get an amplification factor of 3.33, which would be the best and optimal resolution for the $\pm 3\text{V}$ range.

3.4 Characteristics of A/D boards analog voltage output

By controlling the digital input values to output analog voltage, our team obtained a voltage range from $\pm 10\text{V}$ on channel ao0 before saturation occurred. Additionally on Channel ao1 port, our range still remained $\pm 10\text{V}$ until it began to saturate.

4.0 Conclusions

By utilizing the DAQ board as an A/D converter, we were able to test its capabilities even more. When taking in analog input and using LabVIEW to observe what the digital output was, we found which cases followed the Nyquist criterion and which did not. From observing which charts did not match the oscilloscope, we were then able to later understand why this was, which was because our last waveform chart (Fig 3.) had a sampling frequency that was too low for the signal input, resulting in aliasing. This is supported by the idea that the Nyquist criterion for the Fig 3. was not obeyed.

Additionally using the transfer functions in the A/D board allowed our team to determine the amplification factor by recording voltages and noting when saturation occurred at specific frequencies. Understanding at what voltages saturation occurred given our configured voltage in our function generator, it was possible to find out what the amplification factor was defaulted to in the DAQ board (a gain factor of 2).